USA Computing Olympiad

USACO 2026 Third Contest, Silver
Problem 2. Milk Buckets

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Contest has ended.

Submitted; Results below show the outcome for each judge test case

* 1 3.4mb 2ms	* 2 3.4mb 2ms	* 3 3.4mb 2ms	* 4 3.4mb 2ms	* 5 3.4mb 2ms	* 6 3.4mb 14ms	* 7 3.4mb 14ms	* 8 3.4mb 14ms	* 9 3.4mb 14ms	* 10 3.4mb 14ms	* 11 3.4mb 14ms	* 12 8.5mb 344ms
* 13 8.5mb 315ms	* 14 8.5mb 332ms	* 15 8.5mb 346ms	* 16 8.5mb 315ms	* 17 8.5mb 321ms	* 18 8.5mb 338ms	* 19 8.5mb 345ms	* 20 8.5mb 339ms	* 21 8.5mb 330ms	* 22 8.5mb 339ms	* 23 8.5mb 334ms	

English (en)

There are N ($1 \le N \le 2 \cdot 10^5$) buckets in a stack where the i-th bucket from the top has capacity a_i gallons ($1 \le a_i \le 10^9$). A tap above the top bucket sends one gallon of milk per second into the first bucket per second. There is also a pool below bucket N.

When a bucket reaches its capacity after t seconds, at the start of the $t+1$th second it flips to dump its contents into either the bucket below if it is not the last bucket or the pool otherwise (it reverts back to filling at the end of the $t+1$th second). A bucket cannot collect milk while it is flipped; any milk arriving at the bucket above during this second is lost. Additionally, any amount of milk exceeding the capacity of the bucket below is lost.

Handle Q ($1 \le Q \le 3 \cdot 10^5$) queries, each specified by three integers i, v, and t:

1. First, set $a_i = v$ ($1 \le i \le N, 1 \le v \le 10^9$).
2. Then answer the following question: Suppose that at time 0, all buckets as well as the pool are empty. Determine the number of gallons of milk in the pool after t seconds ($1 \le t \le 10^{18}$).

The $a_i = v$ updates persist through later queries.

INPUT FORMAT (input arrives from the terminal / stdin):

The first line contains N.

The second line contains $a_1 \ldots a_N$.

The next line contains Q.

Then the following Q lines contain three integers i, v, and t. This means that you should set $a_i = v$ and then answer the question for t.

OUTPUT FORMAT (print output to the terminal / stdout):

Output the answer to each question on a separate line.

SAMPLE INPUT:

```
3
1 1 1
30
1 1 1
1 1 2
1 1 3
1 1 4

1 1 5
1 1 6
1 1 7
1 1 8
1 1 9
```

```
1 1 10
1 2 1
1 2 2
1 2 3
1 2 4
1 2 5
1 2 6
1 2 7
1 2 8
1 2 9
1 2 10
2 2 1
2 2 2
2 2 3
2 2 4
2 2 5
2 2 6
2 2 7
2 2 8
2 2 9
2 2 10
```

SAMPLE OUTPUT:

```
0
0
0
1
1
2
2
3
3
4
0
0
0
0
1
1
1
2
2
2
0
0
0
0
1
1
1
2
2
2
```

When $a = [1, 1, 1]$,

- Bucket 1 flips at times $2, 4, 6, \ldots$
- Bucket 2 flips at times $3, 5, 7, \ldots$
- Bucket 3 flips at times $4, 6, 8, \ldots$

When $a = [2, 1, 1]$,

- Bucket 1 flips at times $3, 6, 9, \ldots$
- Bucket 2 flips at times $4, 7, 10, \ldots$

- Bucket 3 flips at times $5, 8, 11, \ldots$

When $a = [2, 2, 1]$,

- Bucket 1 flips at times $3, 6, 9, \ldots$
- Bucket 2 flips at times $4, 7, 10, \ldots$
- Bucket 3 flips at times $5, 8, 11, \ldots$

SAMPLE INPUT:

```
2
1 2
10
1 1 1
1 1 2
1 1 3
1 1 4
1 1 5
1 1 6
1 1 7
1 1 8
1 1 9
1 1 10
```

SAMPLE OUTPUT:

```
0
0
0
0
2
2
2
2
4
4
```

SAMPLE INPUT:

```
3
1 1 1
1
1 1 1000000000000000000
```

SAMPLE OUTPUT:

```
499999999999999999
```

SCORING:

- Inputs 4-5: $N \leq 10, Q \leq 100$, and all $t \leq 10^4$
- Inputs 6-11: $N \leq 10^3, Q \leq 10^4$
- Inputs 12-23: No additional constraints.

Problem credits: Akshaj Arora

Contest has ended. No further submissions allowed.

Previous Submissions:

Tue, Feb 24, 2026 04:34:13 EST (C++17)
Tue, Feb 24, 2026 04:41:55 EST (C++17)